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Questions on Markovian Projection and Optimal Control for American Option Pricing

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Markovian Projection

Markovian Projections:

The time evolution of the vector of asset prices $\vec{X} \in \mathbb{R}^d$ is given by:

$$dX_i(t) = a_i(t, X(t))dt + b_{ij}(t, X(t))dW_j(t), \quad 0 \leq t \leq T \quad (1)$$

$$\vec{X}(0) = \vec{x}_0 \quad (2)$$

with $\vec{W}$ a k -dimensional Wiener process with independent components.

Consider the (generally) non-Markovian process $S_1 = \vec{P}_1 \cdot \vec{X}$, $\vec{P}_1 \in \mathbb{R}^{d \times 1}$ with dynamics given by:

$$dS_1(t) = \vec{P}_1 \cdot \vec{a}(t, X(t))dt + \vec{P}_1^T b(t, X(t))d\vec{W}(t), \quad 0 \leq t \leq T \quad (3)$$

$$\vec{S}_1(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (4)$$

We now consider the surrogate Markovian process $\bar{S}^{x_0} \in \mathbb{R}$, whose dynamics are given by:

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \quad t \in [0, T] \quad (5)$$

$$\bar{S}^{x_0}(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (6)$$

and the diffusion coefficients $\bar{a}^{x_0}$ and $\bar{b}^{x_0}$ are determined as:

$$\bar{a}^{x_0}(t, s) = \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, X(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (7)$$

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\vec{P}_1^T \vec{b}(t, X(t)) \vec{b}(t, X(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (8)$$

Key Result: $S_1(t) |_{\vec{X}(0)=\vec{x}_0} = \vec{P}_1 \cdot \vec{X}(t) |_{\vec{X}(0)=\vec{x}_0}$ and $\bar{S}^{x_0}(t)$ have the same conditional distribution for all $t \in [0, T]$.

American Options: Definitions and link to SOC

The price of an American option u_A may be expressed as:

$$u_A(t, x) = \sup_{\tau \in \mathcal{T}_t} \mathbb{E} \left[e^{-r(\tau-t)} g(\vec{P}_1 \cdot \vec{X}(\tau)) \mid \vec{X}(t) = \vec{x} \right], \quad (9)$$

where for any $q \in [0, T]$ the admissible stopping times are given by

$$\mathcal{T}_q = \left\{ \tau : \Omega \rightarrow [q, T] \mid \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in [q, T] \right\}. \quad (10)$$

with the filtration $\mathcal{F}_t = \sigma\{\vec{X}(q) : 0 \leq q \leq t\}$. The American option price may be formulated as a stochastic optimal control (SOC) problem with Hamilton-Jacobi-Bellman (HJB) equation given by:

$$-\partial_t u_A(t, x) = (\mathcal{H}u_A)(t, x), \quad (t, x) \in [0, T] \times D, \quad (11)$$

$$u_A(t, \cdot) = g(\vec{P}_1 \cdot \cdot) \quad (12)$$

The Hamiltonian is written compactly as:

$$(\mathcal{H}u_A)(t, x) = (\mathcal{L}u_A)^+(t, x), \quad (13)$$

with the second order linear differential operator $\mathcal{L}$ given by:

$$(\mathcal{L}v)(t, \vec{x}) = \left(-r + \sum_i a_i \partial_{x_i} + \frac{1}{2} \sum_{i,j} (\mathbf{b}\mathbf{b}^\top)_{ij} \partial_{x_i x_j}^2 \right) v(t, \vec{x}). \quad (14)$$

We now consider the value function $\bar{u}_A(t, s)$ of an American option written on the surrogate underlying process $\bar{S}^{x_0}(t)$, with the dynamics given by (76). $\bar{u}_A$ satisfies the projected HJB equation given by:

$$-\partial \bar{u}_A(t, x) = (\bar{\mathcal{H}}\bar{u}_A)(t, x), \quad (t, x) \in [0, T] \times D, \quad (15)$$

$$\bar{u}_A(t, \cdot) = g(\cdot) \quad (16)$$

The Hamiltonian is written compactly as:

$$(\bar{\mathcal{H}}\bar{u}_A)(t, s) = (\bar{\mathcal{L}}\bar{u}_A)^+(t, s), \quad (17)$$

with the second order linear differential operator $\mathcal{L}$ given by:

$$(\bar{\mathcal{L}}v)(t, s) = \left(-r + \bar{a}^{x_0}(t, s) \partial_s + \frac{1}{2}(\bar{b}^{x_0}(t, s))^2 \partial_{ss}^2 \right) v(t, s). \quad (18)$$

The stopping times of the projected SDE are adapted to the filtration $\bar{\mathcal{F}}_t = \sigma\{\bar{S}^{x_0}(q) : 0 \leq q \leq t\}$ generated by the surrogate process.

Lemma

Let X solve (1),(2) and consider the non-Markovian process $S_1 = P_1 X$. Let

$$d\bar{S}^{(x_0)}(t) = \bar{a}^{(x_0)}(t, \bar{S}^{(x_0)}(t))dt + \bar{b}^{(x_0)}(t, \bar{S}^{(x_0)}(t))d\bar{W}(t), \quad t \in [0, T], \quad \bar{S}^{(x_0)}(0) = P_1 x_0. \quad (19)$$

where $\bar{W}$ is a Wiener process and the drift and diffusion coefficients of the surrogate process $\bar{S}^{(x_0)}$ are determined as follows:

$$\bar{a}^{(x_0)}(t, s) = \mathbb{E}[P_1 a(t, X(t)) \mid P_1 X(t) = s, X(0) = x_0], \quad (20)$$

$$(\bar{b}^{(x_0)}(t, s))^2 = \mathbb{E}[(P_1 b b^\top P_1^\top)(t, X(t)) \mid P_1 X(t) = s, X(0) = x_0]. \quad (21)$$

Then, $S_1(t)|_{\{X(0)=x_0\}} = P_1 X(t)|_{\{X(0)=x_0\}}$ and $\bar{S}^{(x_0)}(t)$ have the same **conditional distribution** for all $t \in [0, T]$.

Proof:

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ and consider the weak approximation error

$$\epsilon_T = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) \mid \bar{S}^{(x_0)}(0) = P_1 x_0] - \mathbb{E}[g(P_1 X(T)) \mid X(0) = x_0]. \quad (22)$$

Let us also define

$$\bar{u}(t, s) = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) \mid \bar{S}^{(x_0)}(t) = s]. \quad (23)$$

Then, we represent the projection weak error as

$$-\epsilon_T = \mathbb{E}[\bar{u}(T, S_1(T)) \mid X(0) = x_0] - \bar{u}(0, S_1(0)) \quad (24)$$

Although S_1 is not Markovian, it is an Ito process which solves

$$dS_1(t) = P_1 a(t, X(t))dt + P_1 b(t, X(t))dW_t, \quad (25)$$

and we have the generator

$$[\mathcal{G}\bar{u}](t, x) = P_1 a(t, x) \partial_s \bar{u}(t, P_1 x) + \frac{1}{2} (P_1 b b^\top P_1^\top)(t, x) \partial_{ss} \bar{u}(t, P_1 x). \quad (26)$$

Then by Ito formula we obtain

$$\bar{u}(T, S_1(T)) = \bar{u}(0, S_1(0)) + \int_0^T [\partial_t \bar{u}(t, S_1(t)) + \mathcal{G}\bar{u}(t, X(t))] dt$$

and consequently,

$$-\epsilon_T = \mathbb{E} \left[\int_0^T (\partial_t \bar{u}(t, S_1(t)) + [\mathcal{G}\bar{u}](t, X(t))) dt \right].$$

Using the fact that $\bar{u}$ solves the projected KBE equation:

$$0 = \partial_t \bar{u} + \bar{\mathcal{G}}\bar{u}, \quad t < T, \quad (27)$$

$$\bar{u}(T, \cdot) = g(\cdot), \quad (28)$$

with coefficients $\bar{a}^{(x_0)}$, $\bar{b}^{(x_0)}$ and

$$\bar{\mathcal{G}}v = \bar{a}^{(x_0)} \partial_s v + \frac{1}{2} (\bar{b}^{(x_0)})^2 \partial_{ss} v,$$

we obtain

$$-\epsilon_T = \int_0^T \mathbb{E}_{0, x_0} \left[\left(P_1 a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1 X(t)) \right) \partial_s \bar{u}(t, P_1 X(t)) \right] dt \quad (29)$$

$$+ \frac{1}{2} \int_0^T \mathbb{E}_{0, x_0} \left[\left((P_1 b b^\top P_1^\top)(t, X(t)) - (\bar{b}^{(x_0)})^2(t, P_1 X(t)) \right) \partial_{ss} \bar{u}(t, P_1 X(t)) \right] dt. \quad (30)$$

Let us now choose $\bar{a}^{(x_0)}$ and $\bar{b}^{(x_0)}$ such that $\epsilon_T = 0$. Using the tower property and that by conditioning on $P_1X(t)$, $\bar{u}(t, P_1X(t))$ and $\bar{a}^{(x_0)}(t, P_1X(t))$ become deterministic functions (so can be taken out of the inner expectation) we obtain:

$$\mathbb{E}_{0,x_0} \left[\left(P_1a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1X(t)) \right) \partial_s \bar{u}(t, P_1X(t)) \right] \quad (31)$$

$$= \mathbb{E}_{0,x_0} \left[\mathbb{E} \left[\left(P_1a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1X(t)) \right) \partial_s \bar{u}(t, P_1X(t)) \mid P_1X(t) \right] \right] \quad (32)$$

$$= \mathbb{E}_{0,x_0} \left[\left(\mathbb{E}[P_1a(t, X(t)) \mid P_1X(t)] - \bar{a}^{(x_0)}(t, P_1X(t)) \right) \partial_s \bar{u}(t, P_1X(t)) \right]. \quad (33)$$

We note that it is precisely because the generators $\mathcal{G}$ and $\bar{\mathcal{G}}$ both end up depending on $\partial_s \bar{u}_A(t, P_1X)$ which allows for factorisation and ensures that the error can be made to vanish for all bounded and continuous functions g (and for all T since nothing depends on our choice of T). A similar derivation applies to the second term. Thus, we conclude that

$$0 = \mathbb{E} \left[g(S_1(T)) - g(\bar{S}^{(x_0)}(T)) \mid X(0) = x_0 \right] \quad (34)$$

for all bounded continuous functions g . Therefore, for all fixed T , the random variable $S_1(T) = P_1X(T)$ has the same conditional distribution as $\bar{S}^{(x_0)}(T)$.

General comments on the problem of pricing American Options and the paper[2] on Markovian Projection

Comments: The difference between the American and the European option pricing problems is that American option problem is an optimal control/ decision problem. We require a strategy $\tau = f(t, x)$, the stopping time, and only then can we evaluate the expectation using Monte Carlo methods/obtain a price like in the European option case. Further, whilst in the European option case we solve a linear PDE in the American case we solve a non-linear PDE (as the Hamiltonian is in general non-linear).

Common approaches to find the price of an American option involve coming up with a preliminary policy τ , then simulating and iteratively improving it. This becomes unsustainable for high-dimensional options due to curse of dimensionality. Further, solving the non-linear PDE in high-dimensions is also subject to curse of dimensionality.

The paper [2] aims to divide the problem of pricing American basket options (a high-dimensional, non-smooth integration problem) into two sub-problems using the method of Markovian projection:

- ① A decision problem in 1D (a low-dimensional, non-smooth integration problem)
- ② Evaluating the coefficients $\bar{a}$ and $\bar{b}$ which involves doing high-dimensional, smooth integrals

Q3.1: Proof of Markovian Projection Error in American Options I

We want to show that:

$$u_A(0, \vec{x}) - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}) \leq \int_0^T \mathbb{E} \left[\mathcal{H} \left(t, \vec{X}(t), \vec{P}_1 D \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1^T \right) - \bar{\mathcal{H}} \left(t, \vec{P}_1 \cdot \vec{X}(t), D \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \right) \right] dt \quad (35)$$

$$\bar{u}_A(0, \vec{P}_1 \cdot \vec{x}) - u_A(0, \vec{x}) \leq \int_0^T \mathbb{E} \left[-H(t, \bar{S}_1(t) \vec{P}_1, Du_A(t, \bar{S}_1(t) \vec{P}_1), D^2 u_A(t, \bar{S}_1(t) \vec{P}_1)) + \bar{H}(t, \bar{S}_1(t), \vec{P}_1 Du_A(t, \bar{S}_1(t) \vec{P}_1), \vec{P}_1^T D^2 u_A(t, \bar{S}_1(t) \vec{P}_1) \vec{P}_1) \right] dt \quad (36)$$

This gives us the bounds on the error of using Markovian projection for American options. We explain below why this error arises in the first place.

Define

$$\bar{u}_A(t, s) = \mathbb{E}[g(\bar{S}(T)) | \bar{S}(t) = s] \quad (37)$$

with a control given by: $\bar{\alpha}(t, x) = \mathbb{1}_{\bar{u}_A(t, P_1 x) > g(P_1 x)}$.

$$u_A(t, \vec{x}) = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(t) = \vec{x}] \quad (38)$$

with a control given by: $\alpha(t, x) = \mathbb{1}_{u_A(t, x) > g(x)}$.

We note that at $t = T$ (expiry date), $\alpha(T, \vec{X}(T)) = \bar{\alpha}(T, \vec{P}_1 \cdot \vec{X}(T)) = 0$ and $\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) = u_A(T, \vec{X}(T)) = g(T, \vec{P}_1 \cdot \vec{X}(T))$.

$$\mathbb{E}[\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(0) = \vec{x}_0] = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(0) = \vec{x}_0] = u_A(0, \vec{x}_0) \quad (39)$$

$$\begin{aligned} u_A(0, x) - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) &= \mathbb{E} \left[\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) \mid X(0) = x_0 \right] - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) \\ &= \mathbb{E}_{0, x_0} \left[\int_0^T dt \, \partial_t \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) + [\mathcal{L} \bar{u}_A](t, x) \right] \\ &= \mathbb{E}_{0, x_0} \left[\int_0^T dt \, -\bar{\mathcal{H}} \left(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \right) \right. \\ &\quad \left. + [\mathcal{L} \bar{u}_A](t, x) \right] \\ &\leq \mathbb{E}_{0, x_0} \left[\int_0^T dt \, -\bar{\mathcal{H}} \left(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \right) \right. \\ &\quad \left. + \mathcal{H} \left(t, \vec{X}(t), \vec{P}_1 D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1^T \right) \right] \quad (40) \end{aligned}$$

Note in the slides there is an error in the inequalities.

The operator $\mathcal{L}v(x, t)$ is given by:

$$\mathcal{L}v(t, x) = \vec{a} \cdot (\vec{P}_1 \partial_x) v(t, x) + \frac{1}{2} \text{Tr}[bb^T (\vec{P}_1 \partial_{xx}^2 v(t, x) \vec{P}_1^T)] \quad (41)$$

In the proof we have used that $\bar{u}_A$ satisfies the HJB equation given by:

$$-\partial_t \bar{u}_A(t, x) = (\bar{\mathcal{H}} \bar{u}_A)(t, x), \quad (t, x) \in [0, T] \times D, \quad (42)$$

$$\bar{u}_A(t, \cdot) = g(\cdot) \quad (43)$$

and that $[\mathcal{L} \bar{u}_A](t, x) \leq \mathcal{H}$. Analogously, we may prove the other inequality (36) using that the error may be re-written in the form:

$$\bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) - u_A(0, \vec{x}) = \mathbb{E} \left[u_A(T, \vec{P}_1 \bar{S}(T)) \mid \bar{S}(0) = \vec{P}_1 \cdot \vec{x}_0 \right] - u_A(0, \vec{P}_1 \bar{S}(0)) \quad (44)$$

where we assume that the basket vector $\vec{P}_1$ is normalised.

Q3.2: Why do the integrands vanish when they take the same decision?

We want to show that the integrands vanish in regions where the original process and the projected one take the same decision. This entails that Markovian projection errors will arise only in the regions where the control processes make different decisions.

- 1 Both in stopping region: both Hamiltonians $\mathcal{H}$ and $\bar{\mathcal{H}}$ in both integrands will be 0 so the error is zero.
- 2 One in stopping region and one in continuation region: then we have the error will, in general, be non-zero since one of the Hamiltonians vanishes but not the other.
- 3 Both in the continuation region: so $\alpha = \bar{\alpha} = 1$.

$$\begin{aligned} I_1 &= \int_0^T dt \mathbb{E}_{0,x_0} [\bar{\mathcal{H}}(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))) \\ &\quad - \mathcal{H}(t, \vec{X}(t), \vec{P}_1 D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1^T)] \\ &= \int_0^T dt \mathbb{E}_{0,x_0} [[\mathcal{L}\bar{u}_A](t, x) - [\bar{\mathcal{L}}\bar{u}_A](t, x)] \underbrace{\quad}_{\text{By our choice of } \bar{a} \text{ and } \bar{b}} = 0 \end{aligned} \tag{45}$$

An analogous proof holds for the other integral/ inequality.

Exact coefficients for Markovian projection

We now show that if we assume that α is the optimal control for u_A , then the projected coefficients must satisfy:

$$\bar{\alpha}^{x_0}(t, s) \bar{a}^{x_0}(t, s) = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1^T \cdot \vec{a}(t, \vec{x}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (46)$$

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (47)$$

Proof: Pricing an American Option involves the optimal control of an SDE using a parameter $\alpha \in \{0, 1\}$. If we assume that alpha is fixed and known then we obtain a standard SDE for the underlying process is simply given by:

$$dS_1(t) = \alpha(t, X(t)) \vec{P}_1^T \cdot \vec{a}(t, \vec{X}(t)) dt + \alpha(t, \vec{X}(t)) \vec{P}_1^T b(t, \vec{X}(t)) d\vec{W}(t), \quad 0 \leq t \leq T \quad (48)$$

$$\vec{S}_1(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (49)$$

With $\alpha(t, \vec{X}(t)) \in \{0, 1\}$. By the Lemma 1 we have that:

$$\bar{\alpha}^{x_0}(t, s) \bar{a}^{x_0}(t, s) = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1^T \cdot \vec{a}(t, \vec{x}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (50)$$

$$(\bar{\alpha}^{x_0}(t, s))^2 (\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\alpha^2(t, \vec{X}(t)) \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (51)$$

Using that $\alpha^2(t, \vec{X}(t)) = \alpha(t, \vec{X}(t))$ and $(\bar{\alpha}^{x_0}(t, s))^2 = \bar{\alpha}^{x_0}(t, s)$ we obtain the required result.

From this we can conclude the Markovian projection error arises because one must project the *combination* of the control $\alpha(t, \vec{X}(t))$ and $\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t))$ (and likewise for the diffusion coefficient). This means that, in general, simply projecting only $\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t))$ and the equivalent diffusion term will not satisfy the above equations and hence not yield the correct coefficients for Markovian projection SDE (correct coefficients are those for which the weak error vanishes). Practically, however, α being unknown at the outset makes it very difficult to satisfy these equations.

Q4: on notation in the paper

We have that the value function for the American option satisfies the following equations:

$$\left\{ \begin{array}{ll} (\mathcal{L}u_A + \partial_t u_A)(t, x) \leq 0, & (t, x) \in [0, T] \times D, \\ u_A(t, x) \geq g(P_1 x), & (t, x) \in [0, T] \times D, \\ (\mathcal{L}u_A + \partial_t u_A)(t, x) (u_A(t, x) - g(P_1 x)) = 0, & (t, x) \in [0, T] \times D. \end{array} \right. \quad (52)$$

Then $\mathcal{H}$ can be written as:

$$(\mathcal{H}u_A)(t, x) = (\mathcal{L}u_A)(t, x) \mathbf{1}_{\max(\mathcal{L}u_A(t, x), u_A(t, x) - g(P_1 x)) > 0} \quad (53)$$

Proof:

One equivalent representation of $\mathcal{H}$ is as follows:

$$\mathcal{H} = \max_{\alpha \in \{0, 1\}} \{\alpha \mathcal{L}u_A\} \implies \mathcal{H} \geq 0 \quad (54)$$

First, this tells us that if and only if $\mathcal{L}u_A \geq 0$ is the $\mathcal{H} = \mathcal{L}u_A$. Hence, the representation using the indicator function must preserve this property. To show that the indicator function preserves this property we do a proof by contradiction. Assume that $\mathcal{L}u_A < 0$ but $u_A(t, x) - g(P_1 x) > 0$ so that the indicator function returns 1. Then by (52)(iii), $\mathcal{L}u_A(t, x) + \partial_t u_A = 0$. But then $\mathcal{L}u_A(t, x) = -\partial_t u_A \geq 0$, which contradicts our initial assumption. The last inequality is implied by the HJB equation and $\mathcal{H} \geq 0$.

Bounds on the American option price

Lower Bound: In the paper, two hitting times for the projected value function $\bar{u}_A$ are defined:

$$\bar{\tau}^* \equiv \inf \left\{ t \in [0, T] : \bar{u}_A \left(t, \bar{S}^{(x_0)}(t) \right) = g \left(\bar{S}^{(x_0)}(t) \right) \right\}, \quad (55)$$

$$\bar{\tau}^\dagger \equiv \inf \left\{ t \in [0, T] : \bar{u}_A \left(t, \vec{P}_1 \cdot \vec{X}(t) \right) = g \left(\vec{P}_1 \cdot \vec{X}(t) \right) \right\}. \quad (56)$$

We also have the hitting time for the value function of the full pricing process u_A given by:

$$\tau^* \equiv \inf \left\{ t \in [0, T] : u_A \left(t, \vec{P}_1 \cdot \vec{X}(t) \right) = g \left(\vec{P}_1 \cdot \vec{X}(t) \right) \right\}. \quad (57)$$

and the class of hitting times $\mathcal{T}_0$:

$$\mathcal{T}_0 = \{ \tau : \Omega \rightarrow [0, T] \mid \{ \tau \leq t \} \in \mathcal{F}_t \text{ for all } t \in [0, T] \}. \quad (58)$$

$\bar{\tau}^* \in \bar{\mathcal{F}}_t$ but $\bar{\tau}^\dagger \in \mathcal{F}_t$. Since u_A is defined as the stopping time which gives the supremum we see that any $\tau \in \mathcal{T}_0$ gives a lower bound to the American option price. Since $\bar{\tau}^\dagger \in \mathcal{T}_0$ we obtain lower bound of the American option price of:

$$u_A(0, x_0) \geq \mathbb{E} \left[\exp(-r \bar{\tau}^\dagger) g \left(\vec{P}_1 \cdot \vec{X}(\bar{\tau}^\dagger) \right) \mid X(0) = x_0 \right]. \quad (59)$$

Upper Bound: Rogers's Dual Representation

Upper Bound: Let $\bar{Z}(t) = e^{-rt} g(\vec{P}_1 \cdot \vec{X}(t))$, let H_0^1 denote the space of all integrable martingales R such that:

$$\mathbb{E}[\sup_{0 \leq t \leq T} |R(t)|] < \infty \text{ and } R(0) = 0 \quad (60)$$

Then the price of the American option is given by:

$$u_A(0, \vec{x}_0) = \inf_{R \in H_0^1} \mathbb{E}[\sup_{0 < t < T} (\bar{Z}(t) - R(t)) | \vec{X}(0) = \vec{x}_0] \quad (61)$$

Proof given in the appendix. The optimising martingale (the martingale reaching the infimum) can be written as:

$$dR^*(t) = e^{-rt} \left((\nabla u_A)^T b \right) (t, X(t)) dW(t), \quad 0 < t < T \quad (62)$$

$$R^*(0) = 0. \quad (63)$$

Finding an optimising martingale can be very complex, so, [2] propose to approximate the optimising martingale $R(t)$ by a near optimal martingale $R^*(t)$:

$$u_A(0, \mathbf{x}_0) \leq \mathbb{E} \left[\sup_{0 \leq t \leq T} \left(\tilde{Z}(q) - R^*(t) \right) \mid X(0) = \mathbf{x}_0 \right], \quad (64)$$

$$dR^*(t) = \exp(-rt) \left((\nabla \bar{u}_A)^T(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1 \cdot \vec{b}(t, X(t)) \right) dW(t), \quad t \in [0, T], \quad (65)$$

$$R^*(0) = 0.$$

General methodology for LSMC:

- 1 Use the dynamic programming principle to reduce the American option pricing problem to many sub-problems (solving for the American option price backwards from expiry at T , first at $t = T - \Delta t$).
- 2 Sample points in space at the time $T - \Delta t$.
- 3 Define a value function via a suitable parametric family of functions. This step is very important as choosing a family of basis functions that poorly approximates will have large impact on the accuracy of the option price. A lot of intuition and insight into the particular problem is required to choose the right parametric family.
- 4 Perform a regression to obtain the option price at $t = T - \Delta t$.
- 5 Compare this "holding value" to that of the early exercise price - when the latter exceeds the former we obtain a stopping time.

Are there cases where LSMC will be obviously better than Markovian projection (MP)?

If you know the right parametric family to use then it is possible but not obvious that LSMC will be better than MP. There are many difficulties with using LSMC including the error propagation

- ① The error from the regression estimate of the continuation value is propagated at each step (backwards). The number of samples you need to alleviate the error propagation further grows exponentially and likewise the number of basis functions required also becomes large.
- ② Markovian projection is a non-convergent method thus giving us limitation on the error that Markovian projection involves.

Essential Dimensionality:

Let $1 \leq n < d$ and $D \subset \mathbb{R}^d$ be a convex set with piecewise smooth boundary. We call a function $v : D \rightarrow \mathbb{R}$ *essentially n -dimensional* if there exist a function $\zeta : \mathbb{R}^n \rightarrow \mathbb{R}$ and a matrix $\mathbf{N} \in \mathbb{R}^{n \times d}$ with orthogonal rows such that $v : \mathbb{R}^d \rightarrow \mathbb{R}$ is given by

$$v(x) = \zeta(\mathbf{N}x).$$

Definition 7.16

By extension, we call a differential operator $\mathcal{K}$ essentially n -dimensional if the following backward PDE is well posed:

$$-\partial_t w(t, x) = \mathcal{K}w(t, x), \quad (t, x) \in [0, T) \times D, \quad w(T, \cdot) = w_T(\cdot), \quad (284)$$

and it has a unique essentially n -dimensional solution for any essentially n -dimensional terminal value w_T .

Here we specifically mean that the function ζ may depend on time, that is,

$$w(t, x) = \zeta(t, \mathbf{N}x),$$

but the matrix $\mathbf{N}$ does not.

Lemma

Separating essential dimensionality and early exercise If a model has a pricing PDE whose differential operator $\mathcal{L}$ is essentially n -dimensional then an American option price is also essentially n -dimensional if the corresponding payoff is essentially n -dimensional in the same subspace.

Proof: We begin by approximating the price of an American option by a Bermudan option that can only be exercised at M equally-spaced intervals: $t_m = m \frac{T}{M}$. Assume that our model has a pricing PDE whose differential operator $\mathcal{L}$ is essentially n -dimensional and that our payoff function $g = g(Px)$.

Base case: consider time interval $t_{M-1} < t < T$. In this time interval, the option is European. By assumption, our option price in this time interval therefore is essentially n -dimensional so may be expressed as:

$$u(t, x) = \bar{u}(t, Px), \quad t_{M-1} < t < T \quad (66)$$

So the value of our option at exercise time t_{M-1} is the maximum of two essentially one-dimensional function:

$$u(t_{M-1}) = \max(u(t_{M-1}^+, x), g(Px)) = \max(\bar{u}(t_{M-1}^+, Px), g(Px)) \quad (67)$$

Now consider an exercise time t_m where we assume that the option price in the interval $t_m < t < t_{m+1}$ is essentially n -dimensional. Then the price of the Bermudan option at a time t_m is given by:

$$u(t_m^-) = \max(u(t_m^+, x), g(Px)) = \max(\bar{u}(t_m^+, Px), g(Px)) = \bar{u}(t_m, Px) \quad (68)$$

From this we can conclude that the price of the Bermudan at all times will be essentially n -dimensional, assuming that it's pricing PDE and the payoff are both essentially n -dimensional on the same subspace.

The American option price is the limit of the Bermudan option price as $M \rightarrow \infty$. Hence, we conclude that given our assumption, the American option price is also essentially n -dimensional.

Relationship between dimensional reduction and exactness of Markovian projection for American Option pricing

Lemma

If a model has a pricing PDE whose coefficients are essentially n -dimensional and the American option price is also essentially n -dimensional on the same sub-space, then the Markovian projection on that n -dimensional subspace is exact, namely it produces exact values of the American option price.

Proof: First we prove that the Hamiltonians corresponding to u_A and $\bar{u}_A$ coincide pointwise for the case $n = 1$ but this proof may be easily extended to higher n , that is:

$$\mathcal{H}(t, s\vec{P}, v\vec{P}, \vec{P}Q\vec{P}^T) - \bar{\mathcal{H}}(t, s, v, Q) = 0 \quad (69)$$

Where we recall that $\mathcal{H}$:

$$\mathcal{H} = \max_{\alpha \in \{0,1\}} (\alpha \mathcal{L}u_A(t, \vec{x})) \quad (70)$$

$$\mathcal{L}v(t, \vec{x}) = (\vec{a}_i \partial_{x_i})v(t, \vec{x}) + \frac{1}{2} (bb^T)_{ij} (\partial_{x_i x_j}^2 v(t, \vec{x})) \quad (71)$$

Assume that the model's pricing PDE has coefficients which are essentially 1-dimensional then:

$$\begin{aligned}\vec{a}(t, \vec{x}) &= \vec{a}'(t, \vec{P} \cdot \vec{x}) \\ b(t, \vec{x}) &= b'(t, \vec{P} \cdot \vec{x})\end{aligned}\tag{72}$$

Assume also that the American option price is essentially 1-dimensional such that $u_A(t, \vec{x}) = \tilde{u}_A(t, \vec{P} \cdot \vec{x})$. We now consider the coefficients of the Markovian projection:

$$\begin{aligned}\bar{a}^{(x_0)}(t, s) &= \mathbb{E}[\vec{P} \cdot \vec{a}(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[\vec{P} \cdot \vec{a}'(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \vec{P} \cdot \vec{a}'(t, s)\end{aligned}\tag{73}$$

$$\begin{aligned}(\bar{b}^{(x_0)}(t, s))^2 &= \mathbb{E}[(\vec{P}^T b b^T \vec{P})(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[(\vec{P}^T b' b'^T \vec{P})(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= (\vec{P}^T b' b'^T \vec{P})(t, s)\end{aligned}\tag{74}$$

From this we can see that the SDE for the true process $S = \vec{P} \cdot \vec{X}$ and the SDE for the projected process $\bar{S}$ will coincide exactly. We can now compare the Hamiltonians:

$$\begin{aligned}\mathcal{H}(t, s\vec{P}, v\vec{P}, \vec{P}Q\vec{P}^T) &= \left(\vec{a}'(t, s) \cdot \vec{P}v + \frac{1}{2} \text{Tr}[(b' b'^T \vec{P}Q\vec{P}^T)] \right)^+, \\ \bar{\mathcal{H}}(t, s, v, H) &= \left(\bar{a}(t, s) v + \frac{1}{2} \bar{b}^2(t, s) Q \right)^+, \end{aligned}\tag{75}$$

Using the expressions for our coefficients derived above we see that these Hamiltonian's coincide pointwise. Since the two Hamiltonians coincide *pointwise* not just in expectation, the error bounds derived previously imply that:

$$u_A(t, \vec{x}) = \bar{u}_A(t, \vec{P} \cdot \vec{x}), \quad \forall (t, x) \in [0, T] \times D.$$

Hence, we conclude that if the American option price is essentially n-dimensional then the Markovian projection on that n-dimensional subspace is exact, i.e. it produces exact values of the American option price.

Lemma (Dimension reduction in the Bachelier model)

Let $X(t)$ solve

$$dX_i(t) = a_i(t, X(t))dt + b_{ij}(t, X(t))dW_j(t), \quad 0 < t < T$$

$$\vec{X}(0) = \vec{x}_0$$

and the drift and volatility given by $a(x, t) = rx$ and $b_B(t, x) = \Sigma$. Furthermore, let $\bar{S}^{(x_0)}(t)$ be the Markovian projection defined by

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \quad t \in [0, T] \quad (76)$$

$$\bar{S}^{x_0}(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (77)$$

and the diffusion coefficients $\bar{a}^{x_0}$ and $\bar{b}^{x_0}$ are determined as:

$$\bar{a}^{x_0}(t, s) = \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, X(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (78)$$

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\vec{P}_1^T \vec{b}(t, X(t)) \vec{b}(t, X(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (79)$$

for $P_1 X(t)$. Then $\bar{S}^{x_0}(t)$ and $P_1 X(t)$ coincide in law.

The Bachelier Model does not feature fat-tailed distribution for returns or clustering of volatility. This issues can be adressed by the introduction of a stochastic clock.

Corollary

Let X follow the multivariate Bachelier dynamics. Let $U(t)$ be an a.s. increasing “stochastic clock” taking values in $\mathbb{R}_+$, independent of X , with $U(0) = 0$. Define the discounted process associated to X by

$$R_X(t) = e^{-rt} X(t). \quad (28)$$

Then the time-changed price process

$$Y(t) = e^{rt} e^{-rU(t)} X(U(t)) = e^{rt} R_X(U(t))$$

has an essentially one-dimensional generator.

Black-Scholes: Dimension Reduction

Aim is to see whether there are parametrisations for which the Black-Scholes Model is essentially one-dimensional. Consider a rotation of the coordinates $\vec{x}$ the B-S equation: $\vec{y} = Q\vec{x}$, $QQ^T = \mathbb{1}$. If the rotated B-S equation takes the same form as the original B-S equation then this entails that B-S is essentially one-dimensional (because we can choose our matrix Q such that we rotate the basket weights from arbitrary $\vec{P}_1$ to $\vec{P} = (1, 0, 0, \dots)^T$).

Black-Scholes in coordinates $\vec{x}$ is given by:

$$-\partial_t w(t, x) = -r w(t, x) + r \sum_i x_i \partial_{x_i} w(t, x) + \sum_{i,j} \Omega_{ij} x_i x_j \partial_{x_i x_j}^2 w(t, x), \quad t \in [0, T], \quad x \in D_{\text{BS}}^d \quad (80)$$

and in rotated coordinates $\vec{y}$:

$$-\partial_t u(t, y) = -r u(t, y) + r \sum_i y_i \partial_{y_i} u(t, y) + \sum_{i,j} \Omega_{ij} (Q^T \vec{y})_i (Q^T \vec{y})_j (Q^T H Q)_{ij} \quad (81)$$
$$H_{mn} = \partial_{y_m y_n}^2 w(t, x)$$

In general, the Black-Scholes model is not essentially one-dimensional. However, under certain particularly choices e.g. where $\Omega_{ij} = C$ it may be shown to be.

Note that in the paper there were minor errors in the derivation of the rotated Black-Scholes but these do not affect final outcome.

3.1.2. *Extrapolation-interpolation to projected domain.* To solve for the projected cost-to-go function, $\bar{u}_A(t, s)$ in (18), we use the Laplace approximation introduced above to evaluate the projected local volatility in a few points in the domain, $\bar{D}$. We extend these values to a truncated domain in which we solve the low dimensional Equation (18). Thanks to the smooth behavior of the projected volatility, $\bar{b}^{(x_0)}$, we only need a relatively low number of evaluations to achieve high accuracy.

However, to verify that the resulting projected cost-to-go function $\bar{u}_A$ is indeed a good approximation of u_A using the lower and upper bounds requires Monte Carlo simulation, which is typically costly compared to the solution of the projected backward problem (44).

Using $\bar{u}_A$ to verify that $\bar{u}_A$ is a good approximation of u_A ?

Although you are using $\bar{u}_A$ to evaluate the bounds you are evaluating $\bar{u}_A(t, \vec{P}_1 \cdot \vec{X})$ i.e. you are using the trajectories of $\vec{X}(t)$, so it is non-circular.

By using $\bar{u}_A$ to estimate the optimising martingale $\bar{R}(t)$ (upper bound) and the stopping time $\bar{\tau}^\dagger$ (lower bound), the authors are loosening the bound. Whatever error we obtain from using these bounds, we know that in reality the error will be smaller.

Laplace approximation vs Regression:

Authors used Laplace approximation as it was fast and gave reasonable approximations for the models they were investigating (Bachelier and Black-Scholes). For more complicated models, Laplace approximation may not work so you need to use regression.

$\mathcal{L}^2$ exercise for estimating coefficients

Comparison of Log Return Distributions for the projected and true process for an equal-weighted basket of 8 stocks

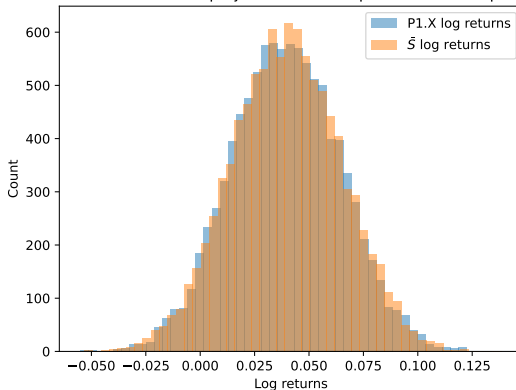


Figure: Histogram comparing the log returns of a basket of 8 stocks with equal weights for the true process and the (Markovian) projected process under Black-Scholes model. $\mathcal{L}^2$ regression was used to estimate the projected volatility coefficient.

Multi-Level Monte Carlo

General Idea of Multi-Level Monte Carlo (MLMC): I

Suppose we have a sequence $P_0, \dots, P_{L-1}$ which approximates P_L with increasing accuracy and cost. Then we can write the expected value of P_L using a telescoping sum:

$$\mathbb{E}[P_L] = \mathbb{E}[P_0] + \sum_{l=1}^L \mathbb{E}[P_l - P_{l-1}] \quad (82)$$

This means that we can use the following unbiased estimator of P_L :

$$Y = N_0^{-1} \sum_{n=1}^{N_0} P_0^{(0,n)} + \sum_{\ell=1}^L \left\{ N_\ell^{-1} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}) \right\}. \quad (83)$$

The total cost and the total variance are then given by (assuming independent samples):

$$\text{Cost} = \sum_{\ell=0}^L N_\ell C_\ell, \quad \mathbb{V}(Y) = \sum_{\ell=0}^L N_\ell^{-1} V_\ell.$$

Assuming a fixed variance, the cost is minimised by choosing N_l to minimise:

$$\sum_{\ell=0}^L (N_\ell C_\ell + \mu^2 N_\ell^{-1} V_\ell),$$

General Idea of Multi-Level Monte Carlo (MLMC): II

with μ^2 a Lagrange multiplier. This gives:

$$N_\ell = \mu \sqrt{\frac{V_\ell}{C_\ell}},$$

Assuming we want an overall variance given by ϵ^2 we require:

$$\mu = \epsilon^{-2} \sum_{\ell=0}^L \sqrt{V_\ell C_\ell},$$

Hence, the optimal total computational cost is given by:

$$C_{\text{total}} = \epsilon^{-2} \left(\sum_{\ell=0}^L \sqrt{V_\ell C_\ell} \right)^2.$$

What is determined by the particular problem is the relative rate at which the variance decreases with level compared to the increase in the computational cost.

Consider case where output on the level P_ℓ is an approximation of the random variable P , which cannot be simulated.

Assuming that Y is an approximation to $\mathbb{E}[P]$, then the Mean Square Error (MSE) is given by:

$$\text{MSE} \equiv \mathbb{E}[(Y - \mathbb{E}[P])^2] = \mathbb{V}[Y] + (\mathbb{E}[Y] - \mathbb{E}[P])^2.$$

The first contribution to the error is the statistical sampling error from using a finite number of Monte Carlo samples and the second is the weak error (bias), which arises from stopping the telescoping sum at level L . We then assume that Y is the multi-level estimator given by:

$$Y = \sum_{\ell=0}^L Y_\ell, \quad Y_\ell = \frac{1}{N_\ell} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}), \quad P_{-1} \equiv 0.$$

then we obtain that assuming independent samples:

$$\mathbb{E}[Y] = \mathbb{E}[P_L], \quad \mathbb{V}[Y] = \sum_{\ell=0}^L \frac{V_\ell}{N_\ell}, \quad V_\ell \equiv \mathbb{V}[P_\ell - P_{\ell-1}].$$

For $MSE < \epsilon^2$, it is sufficient to ensure that each contribution to the MSE is less than $\frac{\epsilon^2}{2}$. Giles (2008) considers a geometric sequence of levels with the cost increasing exponentially and variance decreasing exponentially with level.

Theorem

Let P be a random variable and P_ℓ its level- ℓ approximation. Suppose there exist independent estimators Y_ℓ based on N_ℓ samples, each of cost C_ℓ and variance V_ℓ , and positive constants $\alpha, \beta, \gamma, c_1, c_2, c_3$ with $\alpha \geq \frac{1}{2} \min(\beta, \gamma)$ such that

$$\textbf{Weak Error: } |\mathbb{E}[P_\ell - P]| \leq c_1 2^{-\alpha \ell},$$

$$\mathbb{E}[Y_\ell] = \begin{cases} \mathbb{E}[P_0], & \ell = 0, \\ \mathbb{E}[P_\ell - P_{\ell-1}], & \ell > 0, \end{cases}$$

$$\textbf{Variance: } V_\ell \leq c_2 2^{-\beta \ell},$$

$$\textbf{Cost: } C_\ell \leq c_3 2^{\gamma \ell}.$$

Then for any $\varepsilon < e^{-1}$ there exist L and $\{N_\ell\}$ such that the estimator $Y = \sum_{\ell=0}^L Y_\ell$ satisfies $\mathbb{E}[(Y - \mathbb{E}[P])^2] < \varepsilon^2$, and its cost C obeys

$$\mathbb{E}[C] \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > \gamma, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = \gamma, \\ c_4 \varepsilon^{-2 - (\gamma - \beta)/\alpha}, & \beta < \gamma. \end{cases}$$

Note computational cost for standard Monte Carlo is $\mathcal{O}(\varepsilon^{-2 - \frac{\gamma}{\alpha}})$. (Sketch of proof in Appendix)

We consider SDE's of the form:

$$\begin{aligned}dS_t &= rS_t dt + \sigma S_t dW_t \\ S_0 &= s_0\end{aligned}\tag{84}$$

We then want to compute the expected value of $f(S(T))$, where the function f is assumed for now to be Lipschitz.

A simple Euler-Maruyama discretisation with time step h of the SDE gives:

$$\widehat{S}_{n+1} = \widehat{S}_n + a(\widehat{S}_n, t_n) h + b(\widehat{S}_n, t_n) \Delta W_n.$$

The standard Monte Carlo estimator for $\mathbb{E}[f(S(T))]$ is given by:

$$\widehat{Y} = \frac{1}{N} \sum_{i=1}^N f(\widehat{S}_T^{(i)}).$$

where we implicitly assume a certain step-size h . Assuming that $a(S_t, t)$ and $b(S_t, t)$ satisfy certain conditions then the MSE is given by:

$$MSE \approx c_1 N^{-1} + c_2 h^2\tag{85}$$

Setting $MSE = \epsilon^2$, we require $N = \mathcal{O}(\epsilon^{-2})$ and $h = \mathcal{O}(\epsilon)$. This gives a total computational cost $C_{MC} = \mathcal{O}(\epsilon^{-3})$

Giles' MLMC Method for Euler SDE:

- ① We take a geometric sequence of levels such that at level ℓ the uniform time-step is taken to be $h_\ell = h_0 M^{-\ell}$ with M a positive integer typically taken to be 2.
- ② To couple the coarse and the fine levels we use the same underlying Brownian path and sum the Brownian increments for the fine path time steps to obtain the Brownian increments for the coarse time-steps.
- ③ Assuming a Lipschitz payoff function f (for which for any S_1, S_2 there exists a constant K such that $|f(S_1) - f(S_2)| \leq K\|S_1 - S_2\|$) and that the strong error of the Euler discretisation is $\mathbb{E}[\|S - \hat{S}\|^2] = \mathcal{O}(h)$ that:

$$\begin{aligned}
 V_\ell &\equiv \mathbb{V}[P_\ell - P_{\ell-1}] = \mathbb{V}[(P_\ell - P) - (P_{\ell-1} - P)] \\
 &\leq 2(\mathbb{V}[P - P_\ell] + \mathbb{V}[P - P_{\ell-1}]) \\
 &\leq 2(K^2 \mathbb{E}[\|S - \hat{S}_\ell\|^2] + K'^2 \mathbb{E}[\|S - \hat{S}_{\ell-1}\|^2])
 \end{aligned} \tag{86}$$

This gives that $V_\ell = \mathcal{O}(h_\ell)$. Hence, if we take $M = 2$ we have $\alpha = 1, \beta = 1, \gamma = 1$. This means the complexity to reach a rms error of ϵ^2 is, by Theorem [6], $C_{MLMC} = \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$.

So MLMC is near the optimal lower bound of $\mathcal{O}(\epsilon^2)$ for complexity, save logarithmic factors.

Pricing a European Option under GBM using MLMC

Aim to price a European Call Option on a single stock, with pay-off

$f(S_T) = e^{-rT}(S_T - K)^+$ whose underlying asset satisfies Geometric Brownian Motion (GBM) following the methodology outlined in (Giles 2018).

$$dS_t = rS_t + \sigma S_t dW_t \quad (87)$$

Following (Giles 2018), we choose $r = 0.05$, $\sigma = 0.2$, $S_0 = 100$, $T = 1$, $K = 100$.

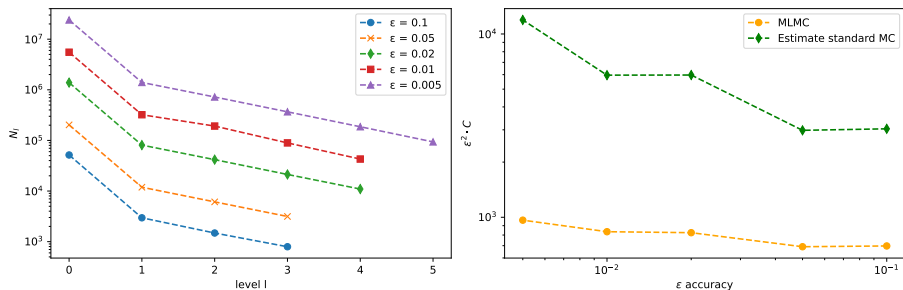


Figure: Left hand figure plots number of paths per level for a selected accuracy ϵ . Right hand figure plots $\epsilon^2 * C$ for MLMC and an estimated $\epsilon^2 * C$ of standard Monte Carlo. Given that the theoretical expected cost is for MLMC for our case is $C \sim \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$ we expect that $\epsilon^2 * C$ should be approximately constant as we see in the plot.

Overview of Weighted Least Square Polynomial Approximation I

Authors consider the case where $f(\mathbf{y})$ is a scalar quantity of interest $Q(u_{\mathbf{y}})$ of the solution $u_{\mathbf{y}}$ to some PDE with parameters $\mathbf{y}$, and where $f_n(\mathbf{y})$ is the corresponding approximation $Q(u_{\mathbf{y},n})$ obtained by solving the PDE with a finite-element solver of maximal element diameter $h := n^{-1}$.

The goal of polynomial weighted least squares (PWLS) is to approximate a function f , $f \in \mathcal{L}^2_{\mu}(\Gamma)$, where μ is a probability measure on parameter space Γ , using out finitely many point evaluations of f . The general methodology goes as follows:

- 1 Choose a finite-dimensional space of polynomials $V \in \mathcal{L}^2_{\mu}(\Gamma)$.
- 2 Compute $f(\vec{y}_i)$ for random samples $(\vec{y}_i)_{i=1}^N$ drawn i.i.d from a distribution ν on Γ , defined by $\frac{d\nu}{d\mu} := \rho$
- 3 Minimise the discrete least squares error:

$$\begin{aligned}\Pi_V f &:= \arg \min_{v \in V} \|f - v\|_{disc}^2 \\ \|f - v\|_{disc}^2 &:= \frac{1}{N} \sum_{j=1}^N w(\vec{y}_j) |f(\vec{y}_j) - v(\vec{y}_j)|^2 := \langle f - v, f - v \rangle_{disc}\end{aligned}\tag{88}$$

where the $w := \rho^{-1}$ is the weight function corrects for the difference between the true distribution μ and the sampling distribution ν .

Overview of Weighted Least Square Polynomial Approximation II

- ④ The coefficients $\vec{c}$ of $\Pi_V f$ with respect to any basis $(B_j)_{j=1}^m$ of V are given by the normal equation:

$$\hat{G}\vec{c} = \vec{v} \quad (89)$$

with $\hat{G}_{ij} = \langle B_i, B_j \rangle_{disc}$ and $v_j = \langle f, B_j \rangle_{disc}$ with $i, j \in \{1, \dots, m\}$ assuming that $\hat{G}$ is invertible. If $\hat{G}$ is not invertible, then we define $\Pi_V f$ as the one with minimal $\mathcal{L}_\mu^2(\Gamma)$ norm.

Since $w\rho = 1$, the semi-norm defined above is a Monte Carlo approximation of the $L_\mu^2(\Gamma)$ norm.

$$\mathbb{E}_\nu[w(Y)g(Y)h(Y)] = \int_\Gamma g(y)h(y)w(y)d\nu = \int_\Gamma g(y)h(y)d\mu = \langle g, h \rangle_{\mathcal{L}_\mu^2} \quad (90)$$

Uniqueness of minimisation requires that $N \geq \dim(V) = m$. If $\nu = \mu$, then $N = \mathcal{O}(m^b \log(m))$ ensures a quasi-optimal polynomial fit, where the specific value of $b \geq 1$ depends on μ . For a **specific optimal distribution** we can obtain the same result with $N = Cm \log(m)$ (Hampton and Doonstan (2015)). This occurs for the weight $w_V^* = \rho_V^{*-1}$ which is associated with density:

$$\rho_V^*(y) := \frac{1}{m} \sum_{j=1}^m |B_j(y)|^2.$$

[6] show that the density of the optimal distribution ν_V^* associated with any downward closed polynomial subspace on $[0, 1]^d$ with respect to the Lebesgue measure $d\lambda$ satisfies:

$$C^{-d} < \frac{d\nu_V^*}{d\lambda} \leq C^d p_d^\infty,$$

with $0 < C < \infty$ and $p_d^\infty(y)$, the Lebesgue density of the d-dimensional arcsine distribution given by:

$$p_d^\infty(y) := \prod_{j=1}^d \frac{1}{\pi \sqrt{y_j(1-y_j)}}.$$

Sampling strategies II

The upper bound implies that the number samples for a near-optimal approximation of f , assuming the weight function associated with the arcsine distribution, is given by $N \simeq C^d m \log(m)$, where C is independent of V (but depends exponentially on d).

The upper bound allows for

- ① Very efficient sampling in many cases (without the need for MCMC methods or acceptance/rejection sampling) from the arcsine distribution
- ② Samples can be drawn from the same distribution for all polynomial subspaces
- ③ Gives a simple weight function independent of V , which is important for adaptive algorithms **Why specifically for adaptive algorithms?**

One way to sample from the univariate arcsine distribution:

- ① Generate a uniformly distributed random variable $X \sim U[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- ② Transform X via: $Y = \frac{1}{2}(1 + \sin(X))$ then Y samples the univariate arcsine distribution.

Multi-Level Weighted Least Squares Polynomial Approximation I

Let $h_l \rightarrow 0$ in the limit $l \rightarrow \infty$ and consider a sequence of functions $f_{h_0}, \dots, f_{h_l}$ which approximate in some suitable sense a function f . We then use the ML estimator to write f as a telescoping sum:

$$f = f_{h_0} + \sum_{l=1}^{\infty} (f_{h_l} - f_{h_{l-1}}) \quad (91)$$

As before, we cut-off this sum at some suitable L . The coarsest approximation f_{h_0} is taken to be of the same order as f and it is cheap to evaluate many samples. $(f_{h_l} - f_{h_{l-1}}) \rightarrow 0$ in the limit $l \rightarrow \infty$ and samples are (increasingly) expensive to evaluate.

Choosing $(h_l)_{l=1}^{\infty}$ to be exponentially decreasing and our sequence of subspaces $(V_l)_{l=1}^{\infty}$ to be exponentially increasing we can write:

$$\mathcal{S}_L(f) := \Pi_{V_L} f_0 + \sum_{l=1}^L \Pi_{V_{L-l}} (f_l - f_{l-1}) \quad (92)$$

where the operator $\Pi_{V_l} : F \rightarrow V_l$, for $0 \leq l \leq L$, is the random weighted least square approximation which projects onto the subspace V_l .

Multi-Level Weighted Least Squares Polynomial Approximation II

Theorem

If $\|f - f_h\|_F \lesssim h^\beta$ and each evaluation of f_h requires work $h^{-\gamma}$, $\forall h > 0$ for some space F for which $\inf_{v \in V_l} \|g - v\|_{L^\infty(\Gamma)} \lesssim \|g\|_F (\dim V_l)^{-\alpha}$, $\forall g \in F$, $l \in \mathbb{N}$, then the multilevel polynomial approximation (with optimal sampling) requires the work

$$W \simeq \varepsilon^{-\max\{\gamma/\beta, 1/\alpha\}} |\log \varepsilon|^r,$$

(with some $r = r(\alpha, \beta, \gamma) > 0$ whose value is known) to achieve

- Convergence in probability:

$$\|S_L(f) - f\|_{L^2_\mu(\Gamma)} \leq \varepsilon \quad (93)$$

with probability $> 1 - \varepsilon^{|\log \varepsilon|}$, $0 < \varepsilon \ll 1$.

- Mean square convergence (if L^∞ bound is available):

$$\mathbb{E} \|\tilde{S}_L(f) - f\|_{L^2_\mu(\Gamma)}^2 \leq \varepsilon^2.$$

Note that the work for single level weighted least squares is $W \simeq \varepsilon^{-\frac{\gamma}{\beta} - \frac{1}{\alpha}} |\log(\varepsilon)|$. ML method therefore reduces the (dominant contribution to) work to **either** the contribution from PDE solutions $\varepsilon^{-\frac{\gamma}{\beta}}$ **or** the contribution from the complexity of polynomial spaces $\varepsilon^{-\frac{1}{\alpha}}$.

- ① If the aim is to compute the projected volatility using this method, what is the weight w for the $\vec{X}(t)$ coming from the SDE?
- ② (Relatedly) In the paper, they use the weight w to optimise (minimise) the number of samples that need to be drawn in our case is there an optimisable weight?
- ③ Would the polynomial subspace also be the Legendre polynomials like in the paper?

Multi-Level Least-Squares Polynomial Approximation of the Diffusion Coefficient $\bar{b}(t, s)$

Black-Scholes Model: Set-up I

We want to estimate:

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\vec{P}_1^T \vec{b}(t, X(t)) \vec{b}(t, X(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (94)$$

Where in the Black-Scholes model we have:

$$\vec{b}(t, X(t))_{ij} = X_i \Sigma_{ij} \quad (95)$$

Using the best mean square approximation we can relate finding an expectation value to an $\mathcal{L}^2$ regression problem:

$$\begin{aligned} \mathbb{E}[X|Y] &= g(Y) \\ g(Y) &= \arg \min_{h \in V} \mathbb{E}[(X - h(Y))^2], \quad V = \{h : \mathbb{E}[h(Y)^2] < \infty\} \end{aligned} \quad (96)$$

Single-Level:

Model coefficients in time and space domain:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E} \left[\left(\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} - h(t, \vec{P} \cdot \vec{X}(t)) \right)^2 | \vec{X}(0) = \vec{x}_0 \right] dt \quad (97)$$

And then apply a Monte Carlo estimator:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \left(\vec{P}^T b(t_n, \vec{X}^m(t_n)) b(t_n, \vec{X}^m(t_n))^T \vec{P} - h(t_n, \vec{P} \cdot \vec{X}^m(t_n)) \right)^2 \quad (98)$$

where $\vec{X}^m(t)$ are independent realisations of $\vec{X}$.

Multi-Level:

We need:

- ① Sequence of exponentially increasing polynomial subspaces $(V_l)_{l=0}^\infty$: We chose subspaces of (total degree) Legendre polynomials which are orthogonal on the interval $[-1, 1]$ with weight function $w = 1$.
- ② Exponentially decreasing time steps $h_0, h_1, \dots, h_l, \dots$: We chose $h_l = h_0 \cdot 2^{-l}$.

Multi-Level Estimator:

$$\mathcal{S}_L(f) := \Pi_{V_L} g_0(\vec{X}^0) + \sum_{l=1}^L \Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) \quad (99)$$

where

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \left\| (g(\vec{X}^l) - g(\vec{X}^{l-1})) - v \right\| \quad (100)$$

where we have purposefully left the argument(s) of v blank. The key difference between the case of random PDEs and this SDE case is that the data is a function of **two** different parameters $\vec{X}^l$ and $\vec{X}^{l-1}$.

Options:

- ① Transport Map
- ② Make $v_l = v_l(X_l, X_{l-1})$ - regression on both fine and coarse paths.
- ③ Stick with simple version we have now that seems to work ok.

Which work? What are the costs? What are the errors?

Scatter Plots for Maximum Total Degree $L = 3$

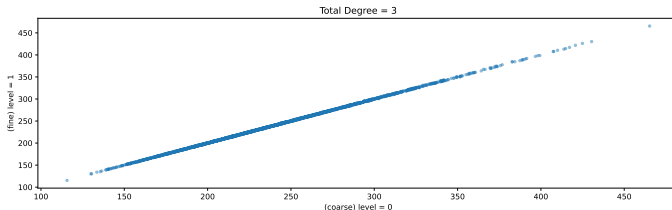


Figure: Sample Pairs from Level 0 and Level 1

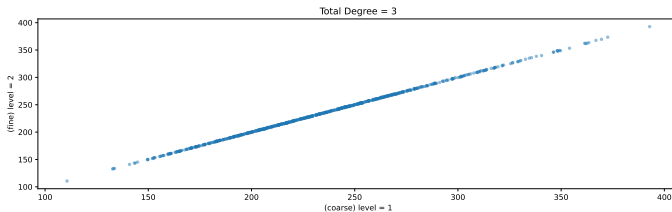


Figure: Sample Pairs from Level 1 and Level 2

Scatter Plots for Maximum Total Degree $L = 3$ II

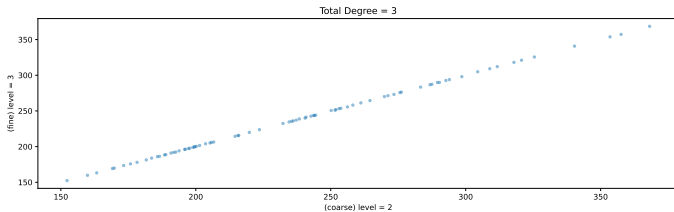


Figure: Sample Pairs from Level 2 and Level 3

Two-Level Example:

Data available:

$$\begin{aligned} \{X_0\}_{m_0=1}^{M_0}, \quad & \{g(X_0)\}_{m_0=1}^{M_0}, \\ \{(X_0, X_1)\}_{m_1=1}^{M_1}, \quad & \{(g(X_0), g(X_1))\}_{m_1=1}^{M_1}. \end{aligned} \tag{101}$$

Goal: Approximate $g(X_1)$ using two levels (particular case):

$$v_0(x) \approx g(x_0), \quad v_{01}(x) \approx g(x_1) - g(x_0).$$

Level 0:

$$v_0 = \arg \min_{v \in V_0} \mathbb{E} \left[(g(X_0) - v(X_0))^2 \right].$$

Level 1:

Determine a transport map:

$$\phi_{10} = \arg \min_{\omega \in W_{10}} \mathbb{E} \left[(X_0 - \phi_{10}(X_1))^2 \right]$$

Carry out $\mathcal{L}^2$ regression using the transport map and only the fine paths X_1 :

$$v_{01} = \arg \min_{v \in V_{01}} \mathbb{E} \left[(g(X_1) - g(\phi_{10}(X_1)) - v)^2 \right].$$

Finally:

$$g(X_1) \approx v_0(\phi_{10}(X_1)) + v_{01}(X_1).$$

Definition of a transport map:

We look for a measurable map

$$T : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{such that} \quad T_{\#}\mu_X = \mu_Y,$$

where μ_X and μ_Y denote the probability measures of X and Y , and $T_{\#}\mu_X$ denotes the push-forward measure. The optimal transport map minimizes a given transportation cost functional. A common choice in the quadratic case is

$$T^* = \arg \min_{T_{\#}\mu_X = \mu_Y} \mathbb{E} \left[|T(X) - X|^2 \right],$$

which, under mild regularity assumptions, admits a unique monotone non-decreasing solution in the one-dimensional setting.

Our SDE case: the measures μ_{h_l} and $\mu_{h_{l-1}}$ are very close in law. We expect therefore that T^* can be written as the identity plus some small displacement.

Implementation of the transport map:

- ① One-dimensional case: Use the Quantile Map

$$T^*(x) = F_Y^{-1}(F_X(x)),$$

which has complexity $\mathcal{O}(N \log N)$ which just comes from sorting the points.

- ② D-dimensional case (our case): we have a few options (we used chatgpt here for suggestions) and becomes more costly.

(1) Gaussian / Affine Brenier map (per time t)

Setup. For each time t , we have samples $X_\ell(t) \in \mathbb{R}^d$ (fine) and $X_{\ell-1}(t) \in \mathbb{R}^d$ (coarse).

Map (quadratic-cost OT between Gaussians). Estimate means/covariances (m_f, Σ_f) from $X_\ell(t)$ and (m_c, Σ_c) from $X_{\ell-1}(t)$, then

$$A_t = \Sigma_c^{-1/2} \left(\Sigma_c^{1/2} \Sigma_f \Sigma_c^{1/2} \right)^{1/2} \Sigma_c^{-1/2}, \quad T_t(x) = m_c + A_t (x - m_f).$$

This is the Brenier map (exact for Gaussians); apply per t .

Complexity per time.

- Moments: $O(n d^2)$ for n samples.
- One eig/SVD: $O(d^3)$; memory $O(d^2)$.

Total over all times N_t : $O(N_t (n d^2 + d^3))$.

We can implement this because in log-space GBM is Gaussian. We aim to compare this to the simple version which seems to work well to see if there is any significant discrepancy between the two.

As theory predicts, using this transport map correctly replicates the diffusion coefficient surfaces.

RMS Error between single level and multi level

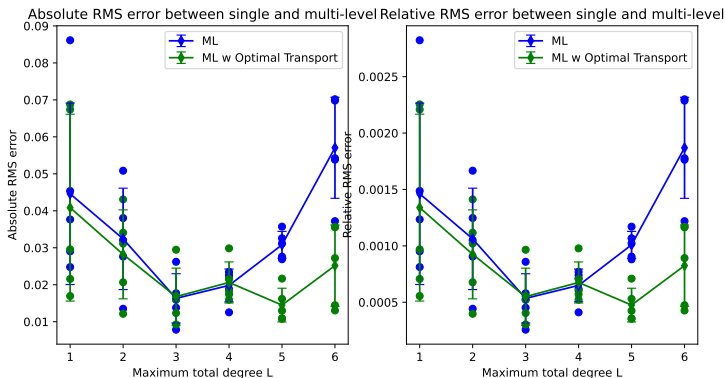


Figure: We take $h_0 = 0.01$, we use $m_l = C(\dim V_l)^2$ regression sample paths.

Possible motivation for the need of a transport map?

Anticipated this might happen as when we plot the diffusion surface for high levels computed via ML plus OT map we do not get the mild non-linearity that we saw when we computed the surface via simple ML.

(3) "Large-scale Optimal Transport map estimation using projection pursuit" Meng et al 2019 I

PPMM Algorithm Idea: Reduce the d-dimensional map estimation to many 1-D problems by projecting samples in the directions which are most "informative" (directions along which X and Y differ most via SAVE dimension reduction) and then applying the simple 1D quantile optimal transport map.

PPMM improves on the **random projection method** which projects vectors into random directions. For random projection, there is no theoretical guide on the number of iterations to convergence and sometimes iterations fail to converge.

Algorithm 1 Select the most "informative" projection direction using SAVE

Input: two standardized matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d}$

Step 1: calculate $\hat{\Sigma} \in \mathbb{R}^{d \times d}$, i.e., the sample variance-covariance matrix of $\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix}$

Step 2: calculate the sample variance-covariance matrices of $\mathbf{X}\hat{\Sigma}^{-1/2}$ and $\mathbf{Y}\hat{\Sigma}^{-1/2}$, denoted as $\hat{\Sigma}_1 \in \mathbb{R}^{d \times d}$ and $\hat{\Sigma}_2 \in \mathbb{R}^{d \times d}$, respectively

Step 3: calculate the eigenvector $\xi \in \mathbb{R}^d$, which corresponding to the largest eigenvalue of the matrix $((\hat{\Sigma}_1 - I_d)^2 + (\hat{\Sigma}_2 - I_d)^2)/4$

Output: the final result is given by $\hat{\Sigma}^{-1/2}\xi/\|\hat{\Sigma}^{-1/2}\xi\|$, where $\|\cdot\|$ denotes the Euclidean norm

Figure: Algorithm to find the most "informative" direction of projection.

(3) "Large-scale Optimal Transport map estimation using projection pursuit" Meng et al 2019 II

Algorithm 2 Projection pursuit Monge map (PPMM)

Input: two matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d}$

$k \leftarrow 0, \mathbf{X}^{[0]} \leftarrow \mathbf{X}$

repeat

(a) calculate the projection direction $\boldsymbol{\xi}_k \in \mathbb{R}^d$ between $\mathbf{X}^{[k]}$ and $\mathbf{Y}$ (using Algorithm 1)

(b) find the one-dimensional OTM $\phi^{(k)}$ that matches $\mathbf{X}^{[k]}\boldsymbol{\xi}_k$ to $\mathbf{Y}\boldsymbol{\xi}_k$ (using look-up table)

(c) $\mathbf{X}^{[k+1]} \leftarrow \mathbf{X}^{[k]} + (\phi^{(k)}(\mathbf{X}^{[k]}\boldsymbol{\xi}_k) - \mathbf{X}^{[k]}\boldsymbol{\xi}_k)\boldsymbol{\xi}_k^\top$ and $k \leftarrow k + 1$

until converge

The final estimator is given by $\hat{\phi} : \mathbf{X} \rightarrow \mathbf{X}^{[k]}$

Figure: PPMM Algorithm

Complexity per time slice for n samples of $\vec{X}$ and $\vec{Y}$:

$$O(Kn d^2 + Kn \log n)$$

where authors find that empirically $K \sim O(d)$.

Simple ML Weighted Least Squares Estimation:

We simulate M_l fine paths $X^f(t^l)$ and construct the coarse paths $X^c(t^{l-1})$ by summing the Brownian increments of the fine paths.

We then have a choice:

- 1 Interpolate the coarse path on the missing fine time steps via a Brownian Bridge.
- 2 Regress the difference only on the coarse time intervals

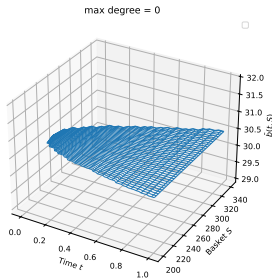
We stick with option 2 for now. We also choose our design matrix or

$$v = v(t_n, \vec{P} \cdot \vec{X}_f^m(t_n^c)).$$

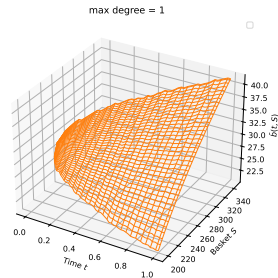
For $l > 0$:

$$\begin{aligned} (\bar{b}_l^{x_0})^2(\cdot, \cdot) - (\bar{b}_{l-1}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{1}{N_{l-1}} \sum_{n=1}^{N_{l-1}} \left([\vec{P}^T b(t_n, \vec{X}_l^m(t_n^{l-1})) b(t_n, \vec{X}_l^m(t_n^{l-1}))^T \vec{P} \right. \\ \left. - \vec{P}^T b(t_n, \vec{X}_{l-1}^m(t_n^{l-1})) b(t_n, \vec{X}_{l-1}^m(t_n^{l-1}))^T \vec{P}] - h(t_n^{l-1}, \vec{P} \cdot \vec{X}_l^m(t_n^{l-1})) \right)^2 \end{aligned} \quad (102)$$

Multi-Level diffusion coefficient surface I

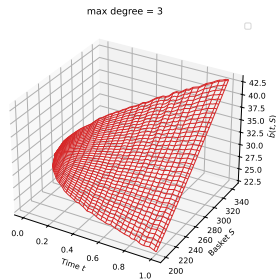
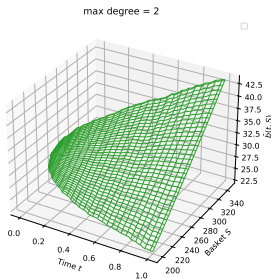


(a) Multi-Level diffusion surface for max total degree = 0



(b) Multi-Level diffusion surface for max total degree = 1

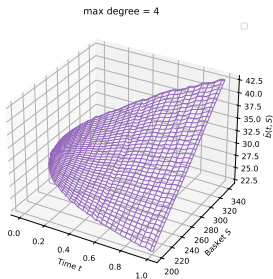
Multi-Level diffusion coefficient surface II



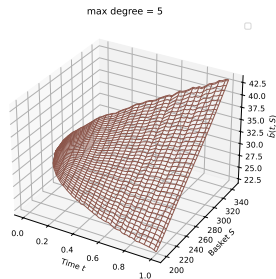
(c) Multi-Level diffusion surface for max total degree = 2

(d) Multi-Level diffusion surface for max total degree = 3

Multi-Level diffusion coefficient surface III

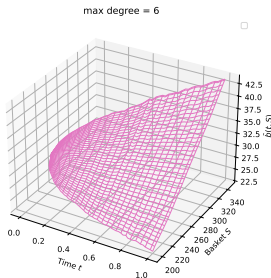


(e) Multi-Level diffusion surface for max total degree = 4

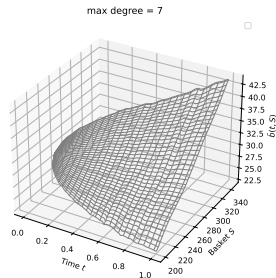


(f) Multi-Level diffusion surface for max total degree = 5

Multi-Level diffusion coefficient surface IV

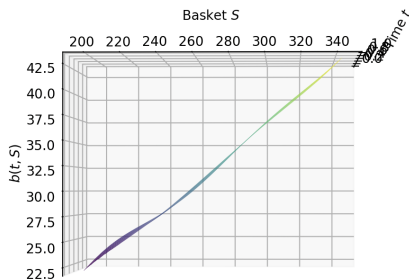


(g) Multi-Level diffusion surface for max total degree = 6. Although it is difficult to observe here we have a mild non-linearity in the $s - \bar{b}$ plane.

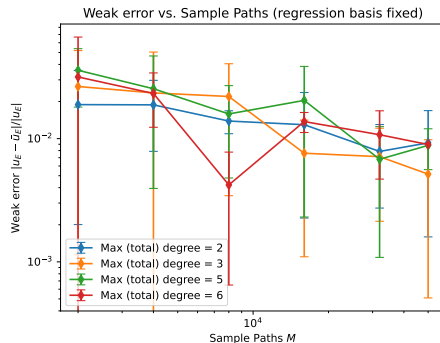


(h) Multi-Level diffusion surface for max total degree = 7. Although it is difficult to observe here we have a mild non-linearity in the $s - \bar{b}$ plane.

Multi-Level diffusion coefficient surface V



(i) Surface for max-degree 7. We observe here a mild non-linearity in the $s - \bar{b}$ plane which is not present in the surfaces for maximum total degree: 2,3,4.

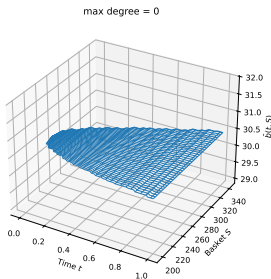


(j) Scaling of the weak-error with sample paths for $\bar{b}(t, s)$ computed with different numbers of levels.

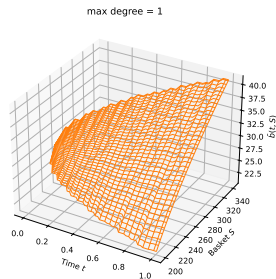
- 1 **Conditioning of the Design Matrix:** if we increase levels we see an increase in conditioning numbers: $k_{l=7} = \mathcal{O}(10^3)$, $k_{l=6} = \mathcal{O}(10^2)$, $k_{l=5} = \mathcal{O}(10^2)$, $k_{l=4} = \mathcal{O}(10^1)$. What conditioning numbers are acceptable? What should/could we do to control the conditioning numbers ridge regression etc?
- 2 When computing the weak error 3 we occasionally have paths which give $\bar{b}^2 < 0$ because they go far out of the high likelihood region. This happens predominantly for higher levels $5 < \cdot$.
- 3 Scaling of (t, s) into the interval $[-1, 1]$ matters - for higher levels we need to take much more conservative scaling factors.

Single Level diffusion coefficient surface I

Parameters: $dt = 0.005$, $M = 400$

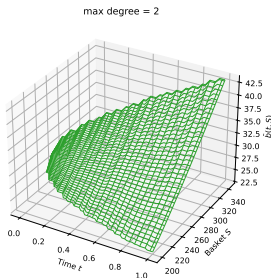


(a) Single-Level diffusion surface for max total degree = 0

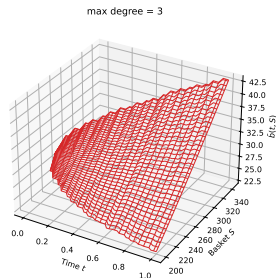


(b) Single-Level diffusion surface for max total degree = 1

Single Level diffusion coefficient surface II



(c) Single-Level diffusion surface for max total degree = 2



(d) Single-Level diffusion surface for max total degree = 3

In the PDE case in the paper they do: To estimate the projection error of our estimate we evaluate the L^2 error norm using Monte Carlo,

$$\|f - S_L(f)\|_{L^2(\Gamma)}^2 \approx \frac{1}{M} \sum_{j=1}^M (f_{L+1}(\mathbf{y}_j) - S_L(f)(\mathbf{y}_j))^2. \quad (103)$$

How do we compute the analogous error in our case?

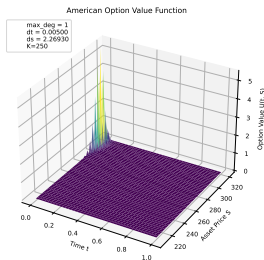
It is not as simple as in the PDE case because we want to compute a conditional expectation. This means we need a proxy for the exact $\bar{b}(t, s)$, presumably, that does not involve projection.

① Laplace Approximation?

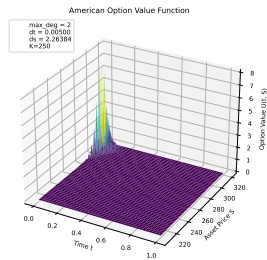
② ...

Want to construct a graph that shows the scaling of the work required to attain a certain accuracy and compare this with the single level, and to see whether this matches the theory predicted by the paper.

Value function for different levels I

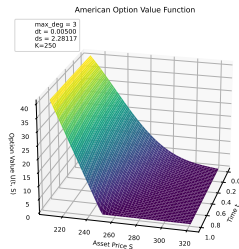


(a) Value function with $\max_{deg} = 1$



(b) Value function with $\max_{deg} = 2$

Value function for different levels II



(c) Value function with $\max_{deg} = 3$

Theorem

Rogers's Dual-Representation of the American Option Price[3]:

Let $\bar{Z}(t) = e^{-rt}g(\vec{P}_1 \cdot \vec{X}(t))$, let H_0^1 denote the space of all integrable martingales M such that:

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} |M(t)|\right] < \infty \text{ and } M(0) = 0 \quad (104)$$

Then the price of the American option is given by:

$$u_A(0, \vec{x}_0) = \inf_{M \in H_0^1} \mathbb{E}\left[\sup_{0 < t < T} (\bar{Z}(t) - M(t)) \mid \vec{X}(0) = \vec{x}_0\right] \quad (105)$$

Proof:

$$\begin{aligned} Y_0 &= \sup_{0 \leq \tau \leq T} \mathbb{E}[Z_\tau] \\ &= \sup_{0 \leq \tau \leq T} \mathbb{E}[Z_\tau - M_\tau] \\ &\leq \mathbb{E}\left[\sup_{0 \leq t \leq T} (Z_t - M_t)\right] \end{aligned} \quad (106)$$

Appendix II

Where we have used the since $M(0) = 0$, $\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = 0$ and for any random variable X_t , $\sup_{\tau} \mathbb{E}[X_\tau] \leq \mathbb{E}[\sup_{\tau} X_\tau]$. Taking the infimum over all $M \in H_0^1$ then proves that Y_0 is bounded by $\mathbb{E}[\sup_{0 \leq t \leq T} (Z_t - M_t)]$.

We use that Y_t has a Doob-Meyer decomposition:

$$Z_t \leq Y_t = Y_0 + M_t^* - A_t^* \quad (107)$$

where M_t^* is a martingale vanishing at 0 and A_t^* is a previsible¹, integrable increasing process also vanishing at 0.

$$\begin{aligned} \inf_{M \in H_0^1} \mathbb{E}[\sup_{0 \leq t \leq T} (Z_t - M_t)] &\leq \mathbb{E}[\sup_{0 \leq t \leq T} (Z_t - M_t^*)] \\ &\leq \mathbb{E}[\sup_{0 \leq t \leq T} (Y_t - M_t^*)] \\ &\leq \mathbb{E}[\sup_{0 \leq t \leq T} (Y_0 - A_t^*)] \\ &= Y_0 \end{aligned} \quad (108)$$

Where we have used that since A_t^* is an increasing process:

$$\sup_{0 \leq t \leq T} (Y_0 - A_t^*) = (Y_0 - A_0^*) = Y_0.$$

Sketch of Giles's proof:

Each of the constraints on the size of the weak error and the statistical error gives a constraint on the number of levels L and the number of samples per level N_ℓ respectively.

- ① $\mathbb{E}[Y - P] = \mathbb{E}[\sum_\ell Y_\ell - P] = \mathbb{E}[P_L - P] \leq c_1 2^{-2\alpha L} < \frac{1}{\sqrt{2}}\epsilon$. This gives that $L \gtrsim \frac{1}{\alpha} \log_2(\frac{1}{\epsilon})$.

- ② We obtain that the optimal N_ℓ is given by: $N_\ell = \frac{2}{\epsilon^2} (\sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}}) \sqrt{\frac{V_\ell}{C_\ell}}$.

The optimal computational cost is then, assuming that $N_\ell \propto 2^{-(\beta+\gamma)\frac{\ell}{2}}$ and $N_\ell C_\ell \propto 2^{(-\beta+\gamma)\frac{\ell}{2}}$:

$$C \lesssim \frac{1}{\epsilon^2} \left(\sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}} \right)^2 \quad (109)$$

We now have three cases depending on the relative magnitudes of β and γ :

- ① $\beta > \gamma$: then we have a sequence that converges so the brackets are constant. Hence $C \lesssim c_4 \epsilon^{-2}$. In this case, most of the computational cost is from the coarse levels where ℓ is low. To further improve the computational cost we can note that adding the further levels will not significantly add to the computational cost. Hence, the equal error split is sub-optimal. Instead, we may give more error to the statistical error/variance term and so fewer samples N_ℓ will be required.
- ② $\beta = \gamma$: then we have $C \lesssim k \epsilon^{-2} (L+1)^2 \approx c_4 \epsilon^{-2} \log(\epsilon)^2$. In this case, computational cost and contribution to the variance is evenly spread across all the levels.
- ③ $\beta < \gamma$: then the dominant contribution to the sum is from level L . So we obtain: $C \lesssim k \epsilon^{-2} 2^{-(\beta-\gamma)L} \approx \epsilon^{-2 - \frac{(-\beta+\gamma)}{\alpha}}$. In this case the dominant computational cost is from the finest levels.

(2) Time-conditioned Brenier map via ICNN

Idea. For quadratic cost, the OT map is $T = \nabla_x \phi$ with ϕ convex. Learn a *single* time-conditioned convex potential $\phi_\theta(t, x)$ (ICNN), and use

$$T_\theta(t, x) = \nabla_x \phi_\theta(t, x).$$

Training objective (sample-based). Aggregate over a set of times $\mathcal{T}$ and minimize a Sinkhorn divergence:

$$\min_{\theta} \sum_{t \in \mathcal{T}} S_\varepsilon\left((T_\theta(t, \cdot))_\# \mu_{X_\ell(t)}, \mu_{X_{\ell-1}(t)}\right),$$

(optionally composed with the affine map from (1) for initialization/stability).

Properties & complexity.

- Monotone (Brenier) map, valid *out-of-sample*, one model for all t .
- Per SGD step: $O(Bp)$ (batch size B , parameters p); no n^2 memory.
- Scales roughly linearly with d ; add small quadratic regularizer to keep T_θ near identity.

(3) Sliced OT with map aggregation (memory-light)

Idea. Reduce to many 1D OT problems and recombine.

Per time t :

- 1 Sample directions $v_1, \dots, v_m \in \mathbb{S}^{d-1}$.
- 2 Project: $u_i^{(k)} = v_k^\top x_i^{(f)}(t)$, $w_j^{(k)} = v_k^\top x_j^{(c)}(t)$.
- 3 Sort $u^{(k)}$ and $w^{(k)}$; match quantiles to get 1D monotone map T_{t,v_k} .
- 4 Aggregate to a d -D map, e.g.

$$T_t(x) \approx \frac{1}{m} \sum_{k=1}^m v_k T_{t,v_k}(v_k^\top x),$$

or fit a small regressor $T_\theta(t, x)$ so that $v_k^\top T_\theta(t, x) \approx T_{t,v_k}(v_k^\top x)$ for all k .

Complexity per time.

$$O(mnd + mn \log n) \quad \text{time}, \quad O(n) \text{ memory},$$

with $m \in [10^2, 10^3]$ typical.

Pros: no n^2 matrices; robust in $d \sim 25$. **Cons:** approximate map.

$$\|f - S_L(f)\|_{L^2(\Gamma)}^2 \approx \frac{1}{M} \sum_{m=1}^M \left(g(\vec{X}_{L+1}^m) - \underbrace{S_L(g)(\vec{P}_1 \cdot \vec{X}_{L+1}^m)}_{\bar{b}_L(t, \vec{P}_1 \cdot \vec{X}_{L+1}^m)} \right)^2. \quad (110)$$

This does not work because $g(\vec{X}_{L+1}^m)$ is not the conditional expectation. This just measures error in projection, which stagnates around ~ 0.04 even as we increase the levels.



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